Chapter 8 Notes

Section 1

Notes

Here is why, for any non-empty class of ordinals B, the smallest ordinal $\alpha \in B$ is an initial element of B. If $\alpha = \emptyset$, then $\alpha \cap B = \emptyset$. If $\alpha \neq \emptyset$, then all members of α are ordinals less than α. But, by assumption, there are no ordinals in B less than α, meaning $\alpha \cap B = \emptyset$.

Exercise 1.1

(1) We first show that R_Ω is well-founded. Consider any non-empty subclass A of R_Ω. Then let α be the lowest rank that any of the elements in A have. Now consider any element $x \in A$ which has rank α. Suppose that there was some element y of A in x. Then y would have rank $\geq \alpha$. But this is not possible, as, by P_6 of Chapter 7 (and the fact that x is a set), y would have a rank $< \alpha$, a contradiction. Thus all elements of x are not in y, meaning $x \cap A = \emptyset$. Thus every element $x \in A$ with rank α is an initial element of A.

(2) The fact that every R_α (where $\alpha > 0$) is well-founded follows because every R_α is a non-empty subclass of R_Ω.

If now it was the case that R_Ω was in fact equal to V, this exercise would prove that V was well-founded.